

SOSNOVO-OZERSKOYE, Russian Federation

WMO: 307450

Lat: 52.538N

Lon: 111.558E

Elev: 3120

StdP: 13.11

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-29.2	-37.6	0.7	-33.1	-33.4	0.9	-28.8	25.2	-0.9	22.5	-0.8	2.8	180	0.729

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	80.6	62.8	77.3	61.7	74.1	60.0	64.9	76.1	63.3	73.5	61.7	71.2	6.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.2	90.6	69.4	59.4	84.9	67.8	57.7	79.9	66.7	31.8	76.0	30.4	73.5	29.2	71.5	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.5	21.3	18.6		DB	-38.7	86.9	6.0	4.3	-43.1	90.0	-46.6	92.5	-49.9	94.9	-54.3	98.0
				WB	-38.9	67.8	5.9	2.0	-43.1	69.2	-46.6	70.4	-50.0	71.5	-54.3	73.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	-9.8	-2.3	12.2	30.0	42.7	57.9	63.0	58.1	44.9	28.7	8.6	-5.4
	DBStd	26.71	10.22	10.55	11.42	8.26	7.37	6.61	4.95	5.54	7.11	8.48	11.56	10.79
	HDD50	9168	1854	1465	1172	600	250	14	0	7	185	660	1243	1718
	HDD65	13731	2319	1885	1637	1049	693	225	97	221	604	1125	1693	2183
	CDD50	966	0	0	0	0	23	250	404	258	31	0	0	0
	CDD65	54	0	0	0	0	0	11	36	7	0	0	0	0
	CDH74	735	0	0	0	0	9	197	406	119	4	0	0	0
	CDH80	121	0	0	0	0	1	31	75	14	0	0	0	0
Wind	WSAvg	7.6	6.6	7.2	7.6	9.1	8.8	7.6	6.9	6.8	8.4	8.3	7.8	6.6
Precipitation	PrecAvg	13.0	0.1	0.1	0.2	0.6	0.9	2.1	3.8	3.1	1.4	0.4	0.2	0.2
	PrecMax	19.4	0.3	0.3	0.6	2.2	3.3	6.6	7.1	6.5	3.3	1.1	0.9	0.4
	PrecMin	7.9	0.0	0.0	0.0	0.0	0.0	0.6	1.1	0.7	0.3	0.0	0.0	0.0
	PrecStd	3.4	0.1	0.1	0.2	0.5	0.7	1.3	1.8	1.2	0.8	0.3	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.8	27.1	45.7	60.6	74.2	83.8	86.0	81.5	73.0	56.1	38.3	22.2
		MCWB	16.3	22.6	35.5	44.2	53.9	62.2	64.2	63.4	56.6	45.2	32.0	20.0
	2%	DB	12.3	21.6	38.5	53.1	67.9	79.0	81.8	77.2	66.8	50.4	33.1	17.2
		MCWB	10.9	18.6	31.7	40.6	50.3	61.1	63.7	61.5	53.9	41.0	28.5	15.4
	5%	DB	8.1	17.5	34.0	47.8	62.5	75.7	78.5	73.5	62.1	45.6	29.2	13.9
		MCWB	6.8	15.2	28.3	37.6	47.9	59.7	62.7	60.0	51.7	37.9	25.8	12.5
	10%	DB	4.3	13.9	30.1	43.6	57.7	72.1	75.2	69.9	57.9	41.2	25.1	10.2
		MCWB	3.3	12.0	25.5	35.0	45.3	58.0	61.7	58.8	49.1	35.0	22.5	9.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.5	22.9	36.0	45.4	55.6	65.8	67.8	65.7	58.3	45.9	32.2	20.1
		MCDB	17.7	27.1	44.1	58.7	71.6	79.5	80.2	75.4	68.1	55.7	37.6	22.0
	2%	WB	11.0	18.7	31.9	41.1	51.6	62.8	65.7	63.6	55.1	41.6	28.7	15.3
		MCDB	12.2	21.6	38.2	52.3	65.7	75.3	78.0	73.0	64.5	49.7	32.9	17.0
	5%	WB	6.9	15.3	28.6	38.0	48.5	60.9	64.2	61.7	52.5	38.3	25.8	12.5
		MCDB	8.1	17.5	33.8	46.8	60.6	72.4	74.9	70.3	60.6	44.7	29.1	13.8
	10%	WB	3.4	12.0	25.4	35.5	45.9	59.1	62.7	59.8	49.8	35.4	22.5	9.0
		MCDB	4.3	13.8	30.0	42.8	57.0	69.9	72.4	68.1	57.6	40.7	25.0	10.1
Mean Daily Temperature Range	5% DB	MDBR	16.3	20.7	23.1	19.7	20.5	20.7	19.9	19.3	19.5	17.3	15.7	15.2
		MCDBR	16.2	21.6	23.5	27.2	31.2	28.9	27.1	27.4	28.3	24.2	17.3	13.5
		MCWBR	15.0	19.1	17.7	17.1	17.3	14.2	12.8	13.6	16.4	16.0	13.9	12.6
	5% WB	MCDBR	16.2	21.4	22.8	25.2	28.5	25.0	21.9	22.5	25.7	22.7	16.8	14.0
		MCWBR	15.1	18.9	17.2	16.4	16.5	12.9	11.3	12.1	15.1	15.4	13.7	13.0
Clear-Sky Solar Irradiance	taub	0.240	0.266	0.300	0.362	0.459	0.445	0.395	0.364	0.325	0.287	0.261	0.241	
	taud	2.194	2.170	2.164	2.187	1.959	2.115	2.333	2.401	2.456	2.465	2.289	2.162	
	Ebn at Noon	242	269	281	275	251	257	270	273	273	263	232	216	
	Edh at Noon	21	29	36	41	55	47	37	33	28	22	19	18	
All-Sky Solar Radiation	RadAvg	400	734	1208	1631	1814	1905	1764	1500	1159	743	423	299	
	RadStd	18	25	33	88	106	122	146	83	82	62	26	16	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	-3.82	-4.50	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)		N/A	N/A	-3.02	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page